

Module 5 Study Notes

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5.0 Introduction

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5.0.1 Why Should I Take this Module?

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We all communicate on networks daily. Looking at social media, streaming video, or researching information on the internet are common activities that we normally don't think much about. However, numerous technological processes are at work to bring us the content that we want from the web.

In this module, you will learn how network protocols work together to allow us to request information and to return that information to us over the network.

5.0.2 What Will I Learn in this Module?

Find the module overview image in the Content folder below:

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Module Title: Network Protocols

Module Objective: Explain how protocols enable network operations.

Topic Title	Topic Objective
Network Communications Process	Explain the basic operation of data networked communications.
Communications Protocols	Explain how protocols enable network operations.
Data Encapsulation	Explain how data encapsulation allows data to be transported across the network.

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5.1 Network Communications Process

Scroll to begin

5.1.1 Networks of Many Sizes

Networks come in all sizes. They range from simple networks that consist of two computers, to networks connecting millions of devices.

Simple home networks let you share resources, such as printers, documents, pictures, and music, among a few local end devices.

Small office and home office (SOHO) networks allow people to work from home, or a remote office. Many self-employed workers use these types of networks to advertise and sell products, order supplies, and communicate with customers.

Businesses and large organizations use networks to provide consolidation, storage, and access to information on network servers. Networks provide email, instant messaging, and collaboration among employees. Many organizations use their network's connection to the internet to provide products and services to customers.

The internet is the largest network in existence. In fact, the term internet means a "network of networks". It is a collection of interconnected private and public networks.

In small businesses and homes, many computers function as both the servers and clients on the network. This type of network is called a peer-to-peer network.

Small Home Networks

Small home networks connect a few computers to each other and to the internet.

Small Office and Home Office Networks

The SOHO network allows computers in a home office or a remote office to connect to a corporate network, or access centralized, shared resources.

Medium to Large Networks

Medium to large networks, such as those used by corporations and schools, can have many locations with hundreds or thousands of interconnected hosts.

World Wide Networks

The internet is a network of networks that connects hundreds of millions of computers world-wide.

5.1.2 Client-Server Communications

All computers that are connected to a network and that participate directly in network communication are classified as hosts. Hosts are also called end devices, endpoints, or nodes. Much of the interaction between end devices is client-server traffic. For example, when you access a web page on the internet, your web browser (the client) is accessing a server. When you send an email message, your email client will connect to an email server.

Servers are simply computers with specialized software. This software enables servers to provide information to other end devices on the network. A server can be single-purpose, providing only one service, such as web pages. A server can be multipurpose, providing a variety of services such as web pages, email, and file transfers.

Client computers have software installed, such as web browsers, email clients, and file transfers applications. This software enables them to request and display the information obtained from the server. A single computer can also run multiple types of client software. For example, a user can check email and view a web page while listening to internet radio.

Examples of common servers:

- File Server - The file server stores corporate and user files in a central location.
- Web Server - The web server runs web server software that allows many computers to access web pages.
- Email Server - The email server runs email server software that enables emails to be sent and received.

Client devices access files with client software such as file explorers; browsers access web servers with browser software; mail clients access email servers with mail client software.

5.1.3 Typical Sessions

A typical network user at school, at home, or in the office, will normally use some type of computing device to establish many connections with network servers. Those servers could be located in the same room or around the world. Below are illustrative scenarios showing how network communications occur in everyday use.

Scenario: Terry (student, BYOD)

Terry uses her cell phone on school Wi-Fi to submit search queries. The device encodes the search terms into binary, sends them as radio waves to the school network, converts to electrical signals on wired segments, traverses ISP networks, and reaches the search engine's servers which respond with results addressed back to Terry's device.

Scenario: Michelle (gamer)

Michelle connects a gaming console via Ethernet to a home network that reaches the ISP by cable modem and router. Gameplay data is sent in many small packets to the game provider's servers and responses (graphics/audio) are returned quickly to enable real-time play.

Scenario: Dr. Awad (medical cloud use case)

Medical imaging (X-rays, MRIs) is digitized and securely transmitted to cloud storage and services. Encryption and secure network services protect patient data while enabling doctors to review, collaborate, and consult remotely.

5.1.4 Tracing the Path

Network traffic often travels through many different networks and points of presence (PoPs), across Tier 1/2/3 ISPs, Internet Exchange Points (IXP), and regional routers. Because routing policies and peering agreements vary, traffic paths can be indirect and differ between outbound and return routes.

Understanding and tracing the path of traffic is essential for cybersecurity analysts to determine true origins and destinations of network communications.

Illustration (concept): multiple clouds (Internet users → Tier 3 → Tier 2 → Tier 1, PoP, IXP) connected by routers and PoPs, showing many possible paths between source and destination.

5.1.5 Lab - Tracing a Route

Lab activities:

1. Verify connectivity to a website (e.g., use `ping` or web browser).
2. Use `tracert` (Linux) or `tracert` (Windows) to examine the network path to a destination host.

3. Use a web-based traceroute tool to compare results and observe intermediate hops.

This lab helps you practice path tracing, interpreting hops, and identifying where latency or routing anomalies occur.

Traceroute Reflections & Architectural Insights

Traceroute is one of the clearest tools for understanding that a network path is not a straight physical line. It is a live conversation between your host and the routers in between. Each probe asks, "How far did you get?" and each router answers by failing on purpose once its countdown timer reaches zero.

1) The Micro View: Physical and Logical Engineering

Traceroute works by sending probes with a low Time-To-Live (TTL), then increasing the TTL one hop at a time. Each router decrements the TTL at Layer 3. When the TTL reaches zero, the router drops the packet and returns an ICMP Time Exceeded (Type 11) message.

That behavior gives you a hop-by-hop map of the path. The key engineering idea is that traceroute never needs to know the full route in advance. It learns the route by forcing each router to reveal itself in sequence.

Direct Line Assumption: One straight, unbroken wire from source to destination.

Traceroute Reality: A series of logical checkpoints, like a postal parcel moving through sorting hubs where each hub stamps its location before forwarding it onward.

2) The Macro View: Global Server Distribution

The same web URL can point to entirely different target IP addresses depending on where you start the test. That happens because large services use Content Delivery Networks (CDNs), mirrored content, and geo-aware load balancing.

Your first hop will not lead to the same edge server as another user's first hop in a different country. The DNS answer varies by region, so your traceroute will show a completely different destination cluster even when the browser name is identical.

The Ticket Analogy: Same airline ticket brand, different city of departure, different physical connection hub, same final service goal.

The Network Reality: Same URL, different origin network, different regional CDN edge, same content delivery outcome.

3) The Routing Policy View: The Non-Shortest Path

Physical geography does not equal network distance. A packet that could travel in a straight geographic line may instead detour through a specific ISP peering hub, a transit provider, or a backbone exchange point because BGP (Border Gateway Protocol) and corporate business policies decide the path.

That means the shortest cable route is not always the shortest network route. Autonomous Systems (ASes) honor routing contracts, peering agreements, and financial path preferences long before they honor physical geography.

The Flight Analogy: A flight between two neighboring small towns often connects through a massive hub city hundreds of kilometers away because the airline network is optimized around centralized hubs.

The Network Reality: Internet traffic follows the provider equivalent of that hub system, even when the destination is physically closer on a geographical map.

Summary Table

View	What You Observe	Why It Happens	Visual Concept
Micro View	TTL decreases and ICMP Time Exceeded reveals each hop.	Routers enforce Layer 3 hop limits and expose the path one step at a time.	A parcel moving through physical sorting checkpoints.
Macro View	The same URL resolves to completely different target IPs.	CDNs and geo-based DNS return nearby edge servers or mirrored content.	Same retail brand, fulfilled by different local warehouses.
Routing Policy	The route looks significantly longer than the physical map suggests.	BGP, peering agreements, and AS boundaries favor business policy over geography.	Commercial airline routes routing through distant connection hubs.

Exam Scenario Checklist

Use this checklist when a word problem mentions traceroute, hops, or path lengths:

- If the problem mentions TTL or countdown clocks: Think Layer 3 processing and ICMP Time Exceeded (Type 11).
- If the exact same website gives different hops/IPs from different terminals: Think CDN optimization or geo-aware DNS routing.
- If the route looks "too long" or moves backwards on a geographical map: Think BGP policy and ISP peering contracts, not physical cable distance.

- If a router appears in the middle of the trace but the destination is still reached: The path is not broken; it is simply longer or differently engineered due to path costs.
- If the trace stops completely and outputs * requests timed out: Suspect firewall security filtering, a missing route entry, or a device configured not to return ICMP messages.

SDN / SD-WAN Perspective (Honours Core Link)

Traditional traceroute reflects the rigid path chosen by completely distributed routing policies (hop-by-hop decisions). SDN and SD-WAN rewrite this rule by adding a centralized control layer that observes these flows globally and steers traffic explicitly.

A software-defined controller can actively favor a lower-latency path, instantly circumvent congested transit hubs, or enforce an application-specific route that completely bypasses default BGP path preferences. In practice, this eliminates jitter, stabilizes voice/video quality, and ensures predictable performance for remote environments.

The Core Architectural Shift: Traditional traceroute shows the rigid path the network chose; SDN allows the application requirements to choose the path dynamically based on real-time latency and policy goals.

Scenario-Based Study Guide: Connecting the Lab to the Protocol Stack

This section ties the theory in 5.2.3 to the practical behavior you observed in Lab 5.1.5. The key idea is that a network request does not move as one thing. It is built, routed, checked, and then stripped back down one layer at a time.

1) Ping and Tracert: What Actually Happens at Layer 3

Ping

- **Layer:** 3 - Network
- **PDU:** Packet
- **Protocol:** ICMPv4 or ICMPv6
- **Action:** Your host sends an **ICMP Echo Request** inside an IPv4 or IPv6 packet to the destination IP address.
- **Inbound result:** If the target is reachable, it returns an **ICMP Echo Reply**.
- **Lab significance:** Ping is a quick reachability test, not an application test. If ping fails, the problem may be routing, filtering, address resolution, or the remote host itself. If ping succeeds, the path is at least working at Layer 3.

Tracert / traceroute

- **Layer:** 3 - Network
- **PDU:** Packet
- **Protocol:** **ICMPv4** or **ICMPv6** with **TTL / Hop Limit** control
- **Action:** The source sends probes with a low **TTL** value. Each router on the path decrements the TTL by 1.
- **Router behavior:** When TTL reaches 0, the router discards the packet and sends back an **ICMP Time Exceeded** message to the source.
- **Inbound result:** The source learns one hop at a time and can map the path to the destination.
- **Lab significance:** This is why tracert shows hop numbers. Each hop is a router that returned a time-exceeded response because the packet was intentionally stopped before the destination.

Why TCP is not used for ping or tracert probes

- **TCP** is designed for reliable application delivery, not for simple path discovery.
- **ICMP** is lighter and gives direct control messages such as Echo Reply and Time Exceeded.
- In your lab, a delay at hop 5 does not mean the web session is broken. It means that router or network segment is slow to respond. If the final destination still works, **TCP** can still recover by waiting, sequencing, and retransmitting where needed, while **UDP** would not repair lost application data.

2) nslookup: The DNS Tool at Layer 7

- **Layer:** 7 - Application
- **PDU:** Data
- **Protocol:** **DNS**
- **Action:** **nslookup** sends a query to a DNS server asking for a name-to-address mapping, usually an **A** record for IPv4 or an **AAAA** record for IPv6.
- **Inbound result:** The DNS server returns the resolved IP address, alias chain, or an error such as NXDOMAIN.
- **Lab significance:** If **nslookup** succeeds but ping fails, name resolution is working and the problem is farther down the stack. If **nslookup** fails, the issue is usually DNS server reachability, configuration, or name resolution itself.

3) ARP: The Missing Link Before Ping Reaches the Internet

- **Layer:** 2 - Data Link / Network Access

- **PDU:** Frame-level local delivery support
- **Protocol: ARP**
- **Action:** Before your IPv4 ping can leave the local LAN, your host must know the MAC address of the default gateway. **ARP** resolves that IPv4 address to a hardware address.
- **Lab significance:** If ARP fails, the ping never reaches the router. That is why a local gateway test may fail even though the IP configuration looks correct. In other words, ARP is the bridge between the Layer 3 destination and the Layer 2 frame that can actually leave your NIC.

4) Terry (Student, BYOD): Layer-by-Layer Technical Timeline

Outbound journey

Step	OSI Layer	Protocols	What Terry's device is doing	Why it matters in the lab
1	Layer 1	Radio / electrical signaling	Her phone or laptop turns bits into Wi-Fi signals	If the signal is weak, you may see retries, slow responses, or dropped hops
2	Layer 2	WLAN	It joins the school SSID and forms a local wireless frame	A failed association means nothing higher can work
3	Layer 7	DHCPv4 or DHCPv6 / SLAAC	The device gets an IP address, gateway, and DNS info	Without this, ping, tracert, and DNS will all fail or behave inconsistently
4	Layer 7	DNS	Terry enters a search query and the browser resolves the search engine name	This is the same logic you tested with nslookup
5	Layer 2	ARP	The device resolves the MAC address of the default gateway	This is the missing link that lets the first packet leave the local network
6	Layer 4	TCP	The browser opens a reliable session for the search request	TCP is chosen over UDP because web traffic must arrive intact and in order
7	Layer 3	IPv4 or IPv6	Packets are addressed and routed toward the internet	The path may cross multiple routers, ISPs, and IXPs
8	Layer 3	ICMP during troubleshooting	Ping and tracert can verify the path or reveal a failing hop	A high-latency hop may show congestion, not necessarily total outage
9	Layer 2	Ethernet/WLAN	Frames move between the laptop, access point, router, and upstream network	Your lab hops represent these Layer 3 forwarding devices between Layer 2 links

Inbound journey

Step	OSI Layer	Protocols	What returns to Terry	Lab meaning
1	Layer 3	ICMP Echo Reply or regular IP return traffic	The destination or an intermediate router sends a response back	Ping success proves reachability; tracer hop replies prove path visibility
2	Layer 4	TCP	Search page data is reassembled reliably	Delay at one hop increases wait time, but TCP can recover if the path still exists
3	Layer 3	IPv4 or IPv6	The response is routed back across the internet	Return paths can differ from outbound paths, which is why hops may not mirror perfectly
4	Layer 2	WLAN	The response reaches Terry's access point and device	Wireless loss or retransmission can inflate latency in lab measurements
5	Layer 7	HTTPS	The browser decrypts and renders the search results	If HTTPS works but ping fails, the issue may be ICMP filtering rather than application failure

5) Troubleshooting Cheat Sheet

Symptom	Most likely OSI layer	What to check first	Lab interpretation
Request timed out	Layer 3, sometime s Layer 4	IP address, default gateway, router path, ICMP filtering	If ping fails but nslookup works, the name is fine and the path is the issue
Destination host unreachable	Layer 2 or Layer 3	ARP , gateway MAC, local subnet, routing table	The packet may not be leaving the LAN or the router has no path onward
DNS server not responding	Layer 7	DNS settings, server reachability, firewall, resolver	Name resolution is broken, so the browser cannot even find the target IP
TTL expired in transit	Layer 3	Routing loop, path instability, overloaded router	A router sent ICMP Time Exceeded because the probe died before reaching the destination
Ping works but web fails	Layers 4 through 7	TCP, HTTPS , proxy, TLS, firewall	The network path exists, but the application session is being blocked or reset
One hop shows high latency	Layer 3	That router, WAN link, or peering point	High latency at one hop may be normal ICMP deprioritization or a real bottleneck; confirm with repeated tests
ARP resolves, but nothing leaves the network	Layer 3	Default gateway, routing, ACLs, NAT	The local link is fine, but forwarding or policy enforcement may be stopping the packet

6) Practical Rule for Reading Your Lab Results

- If **nslookup** fails, start at **Layer 7**.
- If **nslookup** succeeds but **ping** fails, start at **Layer 3** and then check **Layer 2** if the failure is local.
- If **ping** works but **tracert** stops at a router, look for routing, ACL, or ICMP filtering issues at **Layer 3**.
- If the page loads slowly but the path is reachable, suspect congestion or retransmission effects at **Layer 4**.

Lessons Learned & Performance Review

Executive Summary

You show strong command of **data flow** and **PDU identification**. You consistently recognized how information moves across the stack, and you correctly identified the major delivery behaviors for **flow control**, **UDP best-effort transport**, **broadcast delivery**, **segment framing at the Transport layer**, and **layer isolation** as a real-world benefit of the layered model. Your current strength is not in the basics of “what is a network,” but in understanding how traffic is packaged, moved, and interpreted across layers.

Deep Dive Correction Table

The Question	My Initial Logic (The Misconception)	The Correct Technical Reality
In your earlier lab, you could ping 8.8.8.8 but could not resolve "www.cisco.com". At which OSI layer did communication fail when you tried to use the domain name?	I chose the Network layer because ping to an IP address worked, so I assumed the problem had to be routing.	The failure is at Layer 7 - Application because DNS performs name resolution. If IP-based ping works but the domain name does not, the Network layer is functioning and the name lookup function is failing.
A web server receives bits from a client. After converting the bits back into a frame, what is the very next piece of information the server will look at to see if the data is intended for its hardware?	I focused on the destination IP address, assuming software would check that first.	The first check is the destination MAC address at Layer 2 - Data Link . The NIC uses the frame header to decide whether to accept the frame before the packet is processed at Layer 3.
If a sender transmits data faster than a receiver can process it, which timing component is responsible for negotiating a slower speed to prevent data loss?	I was unsure, but I knew it had to do with timing and speed.	Flow control is the Layer 4 - Transport timing function that regulates the rate of delivery so the receiver is not overwhelmed.
During encapsulation, what is the correct name for the PDU after the Transport layer adds port numbers but before the Network layer adds IP addresses?	I recognized that the data was chopped into chunks, but I did not tie the name to the correct layer.	The PDU is a segment at Layer 4 - Transport . This layer adds port numbers and prepares the data for reliable end-to-end communication.
A network administrator notices a broadcast storm where every device on a LAN is receiving the same message simultaneously. Which delivery method is being used?	I knew it was broadcast, but I did not connect it to the delivery model precisely.	Broadcast is a Layer 2 - Data Link delivery method that sends one message to all hosts on the local network segment.
Which layer is responsible for "dressing" the data in encryption	I first chose the Application layer because the	Layer 6 - Presentation is responsible for encryption, compression, and data formatting. It

The Question	My Initial Logic (The Misconception)	The Correct Technical Reality
like HTTPS to ensure that it remains gibberish if intercepted?	browser starts the request there.	prepares the information so it can be safely interpreted by the receiving host.
In the TCP/IP suite, which protocol is considered connectionless and best-effort, meaning it does not confirm if the data actually arrived?	I answered correctly with UDP.	UDP is the Transport layer protocol that sends datagrams without establishing a reliable session. It prioritizes speed and low overhead over delivery confirmation.
What happens to a frame that is received by a network interface card but is smaller than the minimum size required by the protocol?	I answered correctly that it should be dropped.	The frame is rejected at Layer 2 - Data Link because Ethernet enforces minimum frame size rules. Frames that do not meet the protocol requirements are discarded by the NIC.
When a router receives a packet and looks at the header to decide which path to take to reach a destination across the internet, which layer is it operating at?	I thought routing was a Transport-layer function because the packet still needed to move end to end.	Routing is a Layer 3 - Network function. Routers use IP addresses to make forwarding decisions and select the next hop across networks.
One benefit of a layered model is Layer Isolation. Which scenario best describes this benefit?	I answered correctly that changing Ethernet to Wi-Fi should not force a browser reinstall.	Layer isolation means one layer can change without forcing redesigns in adjacent layers. In your example, the Layer 1/2 access method changes, but the browser at Layer 7 does not need to be reinstalled.

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+### Key Differentiators Cheat Sheet

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+ - **DNS vs. Ping:** **DNS** belongs to **Layer 7 - Application** because it resolves names into addresses. **Ping** uses **ICMP** at **Layer 3 - Network** because it tests reachability to an IP address and confirms whether packets can traverse the routed path.

+ - **MAC vs. IP Addressing:** During de-encapsulation, the hardware checks the **MAC address** first at **Layer 2 - Data Link** so the NIC knows whether the frame is intended for it. The **IP address** is checked next at **Layer 3 - Network** so the host can determine whether the packet is meant for that device at the logical addressing level.

+ - **Routing vs. Transport:** **Routing** is a **Layer 3 - Network** responsibility because routers choose the path using IP addresses and forwarding tables. **Transport** is a **Layer 4 - Transport** responsibility because TCP and UDP manage ports, sessions, reliability, and delivery behavior between endpoints.

+ - **Hardware vs. Software Order:** Hardware handles the frame first, software interprets the packet next, and the application only sees the data after the lower-layer headers have been removed. That order is why a local link problem can break connectivity before a browser ever sees an error message.

+ - **Presentation vs. Application:** The Application layer creates the request, but the Presentation layer makes it usable and secure through encoding, encryption, and formatting. If you confuse the two, remember that the browser decides what to ask for, while the Presentation layer decides how that information is packaged safely.

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+### Strategic Focus

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+ - Review **Layers 3 through 7** with special attention to where **DNS, encryption, addressing, and routing** actually occur.

+ - Rehearse the difference between **MAC addresses** and **IP addresses** until you can explain the order of checks during de-encapsulation without hesitation.

+ - Practice identifying when a problem is **Application-layer** name resolution versus **Network-layer** reachability.

+ - Revisit **router behavior** so you can distinguish packet forwarding at **Layer 3** from port-based reliability and retransmission at **Layer 4**.

+ - Continue drilling **PDU names: data, segment, packet, frame, and bits**.

+ - Use lab results from **ping, tracert, and nslookup** to map symptoms back to the correct OSI layer before choosing a fix.

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+Notes and further study suggestions:

- Review how protocols at different layers (physical, data link, network, transport, application) interact during typical sessions.

- Practice using network utilities: `ping`, `tracert` / `tracert`, `nslookup` / `dig`, and packet capture tools like Wireshark.
 - Consider lab scenarios for client-server interactions, secure data transfer (TLS), and routing behavior across ISPs.
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5.2 Communications Protocols

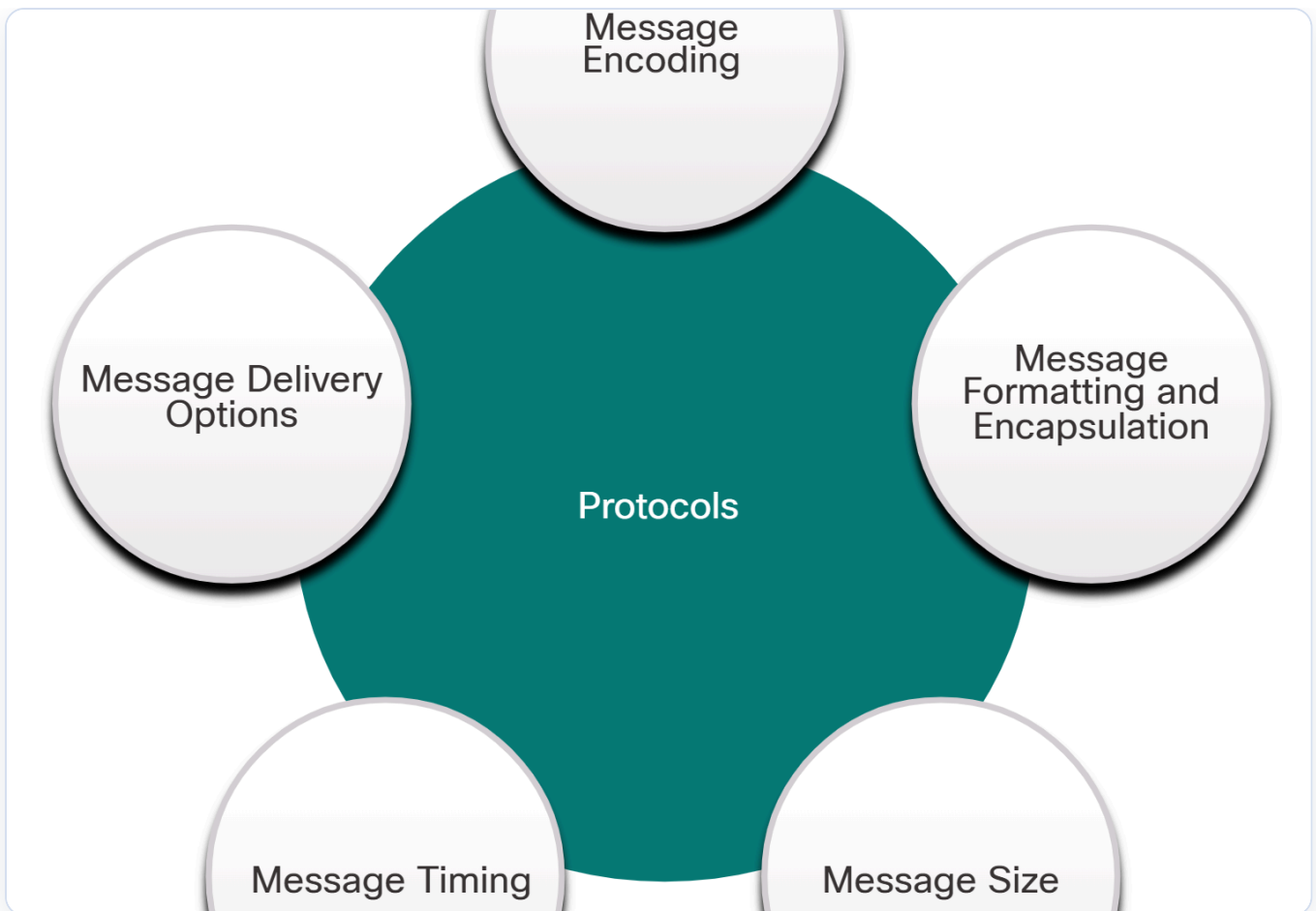
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5.2.1 What are Protocols?

Simply having a wired or wireless physical connection between end devices is not enough to enable communication. For communication to occur, devices must know "how" to communicate. Communication, whether by face-to-face or over a network, is governed by rules called protocols. These protocols are specific to the type of communication method occurring.

For example, consider two people communicating face-to-face. Prior to communicating, they must agree on how to communicate. If the communication is using voice, they must first agree on the language. Next, when they have a message to share, they must be able to format that message in a way that is understandable. For example, if someone uses the English language, but poor sentence structure, the message can easily be misunderstood.

Similarly, network protocols specify many features of network communication:



Key protocol characteristics:

- Message encoding
- Message formatting and encapsulation
- Message size
- Message timing
- Message delivery options

5.2.2 Network Protocols

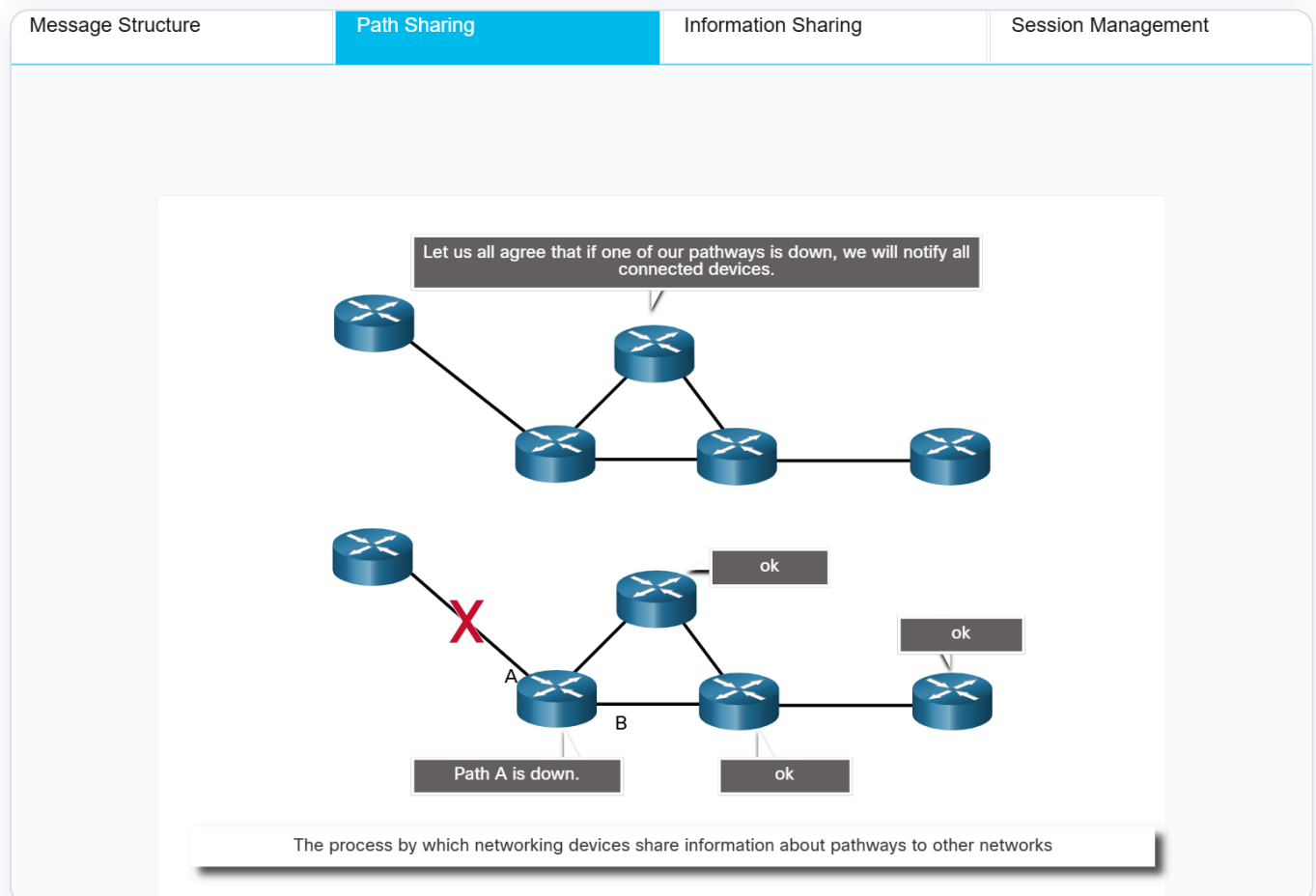
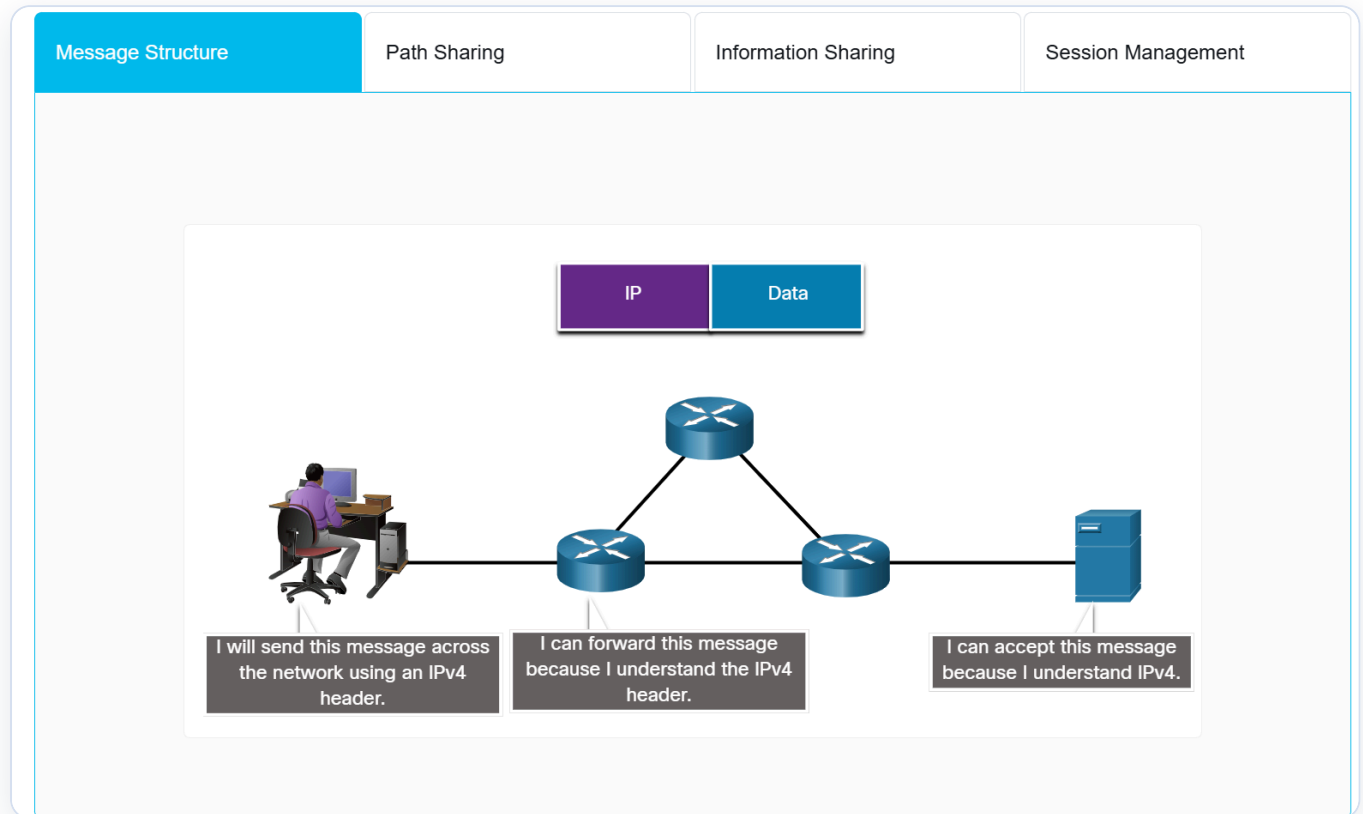
Network protocols provide the means for computers to communicate on networks. Network protocols dictate the message encoding, formatting, encapsulation, size, timing, and delivery options. Networking protocols define a common format and set of rules for exchanging messages between devices.

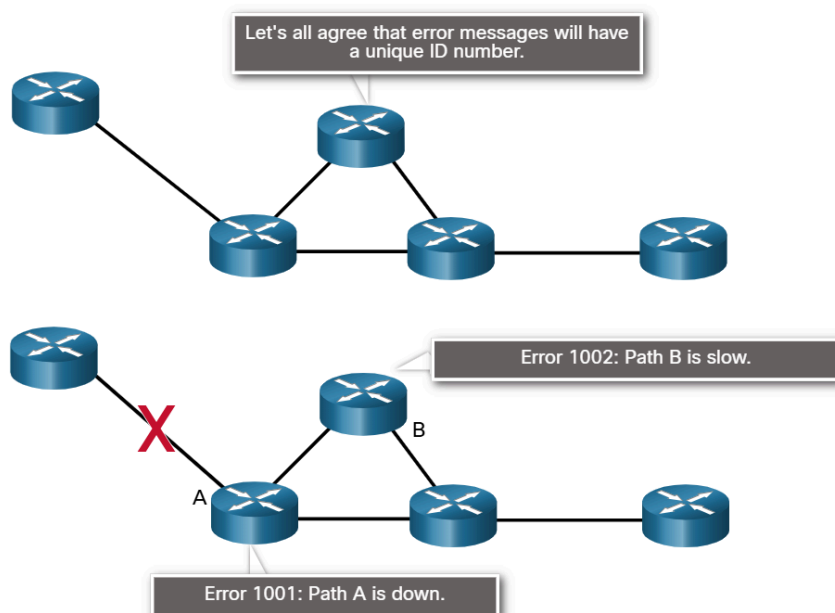
Some common networking protocols are:

- **HTTP** - Hypertext Transfer Protocol
- **TCP** - Transmission Control Protocol
- **IP** - Internet Protocol

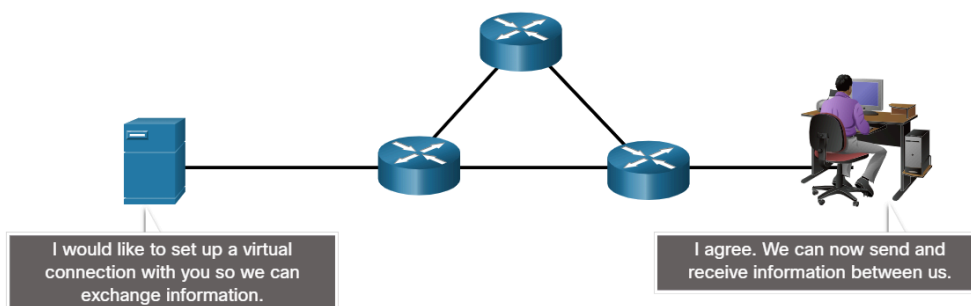
Note: IP in this course refers to both the IPv4 and IPv6 protocols. IPv6 is the most recent version of IP and will eventually replace the more common IPv4.

As a cybersecurity analyst, you must be very familiar with the structure of protocol data and how the protocols function in network communications.





How and when error and system messages are passed between devices



The setting up and termination of data transfer sessions

5.2.3 The TCP/IP Protocol Suite

Today, the TCP/IP protocol suite includes many protocols and continues to evolve to support new services.

Application Layer

Name System

- **DNS** - Domain Name System. Translates domain names such as cisco.com into IP addresses.

Host Configuration

- **DHCPv4** - Dynamic Host Configuration Protocol for IPv4. A DHCPv4 server dynamically assigns IPv4 addressing information to DHCPv4 clients at start-up and allows the addresses to be re-used when no longer needed.
- **DHCPv6** - Dynamic Host Configuration Protocol for IPv6. DHCPv6 is similar to DHCPv4. A DHCPv6 server dynamically assigns IPv6 addressing information to DHCPv6 clients at start-up.
- **SLAAC** - Stateless Address Autoconfiguration. A method that allows a device to obtain its IPv6 addressing information without using a DHCPv6 server.

Email

- **SMTP** - Simple Mail Transfer Protocol. Enables clients to send email to a mail server and enables servers to send email to other servers.
- **POP3** - Post Office Protocol version 3. Enables clients to retrieve email from a mail server and download the email to the client's local mail application.
- **IMAP** - Internet Message Access Protocol. Enables clients to access email stored on a mail server as well as maintaining email on the server.

File Transfer

- **FTP** - File Transfer Protocol. Sets the rules that enable a user on one host to access and transfer files to and from another host over a network. FTP is a reliable, connection-oriented, and acknowledged file delivery protocol.
- **SFTP** - SSH File Transfer Protocol. As an extension to Secure Shell (SSH) protocol, SFTP can be used to establish a secure file transfer session in which the file transfer is encrypted. SSH is a method for secure remote login that is typically used for accessing the command line of a device.
- **TFTP** - Trivial File Transfer Protocol. A simple, connectionless file transfer protocol with best-effort, unacknowledged file delivery. It uses less overhead than FTP.

Web and Web Service

- **HTTP** - Hypertext Transfer Protocol. A set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the World Wide Web.

- **HTTPS** - HTTP Secure. A secure form of HTTP that encrypts the data that is exchanged over the World Wide Web.
- **REST** - Representational State Transfer. A web service that uses application programming interfaces (APIs) and HTTP requests to create web applications.

Transport Layer

Connection-Oriented

- **TCP** - Transmission Control Protocol. Enables reliable communication between processes running on separate hosts and provides reliable, acknowledged transmissions that confirm successful delivery.

Connectionless

- **UDP** - User Datagram Protocol. Enables a process running on one host to send packets to a process running on another host. However, UDP does not confirm successful datagram transmission.

Internet Layer

Internet Protocol

- **IPv4** - Internet Protocol version 4. Receives message segments from the transport layer, packages messages into packets, and addresses packets for end-to-end delivery over a network. IPv4 uses a 32-bit address.
- **IPv6** - IP version 6. Similar to IPv4 but uses a 128-bit address.
- **NAT** - Network Address Translation. Translates IPv4 addresses from a private network into globally unique public IPv4 addresses.

Messaging

- **ICMPv4** - Internet Control Message Protocol for IPv4. Provides feedback from a destination host to a source host about errors in packet delivery.
- **ICMPv6** - ICMP for IPv6. Similar functionality to ICMPv4 but is used for IPv6 packets.
- **ICMPv6 ND** - ICMPv6 Neighbor Discovery. Includes four protocol messages that are used for address resolution and duplicate address detection.

Routing Protocols

- **OSPF** - Open Shortest Path First. Link-state routing protocol that uses a hierarchical design based on areas. OSPF is an open standard interior routing protocol.
- **EIGRP** - Enhanced Interior Gateway Routing Protocol. An open standard routing protocol developed by Cisco that uses a composite metric based on bandwidth, delay, load and reliability.

- **BGP** - Border Gateway Protocol. An open standard exterior gateway routing protocol used between Internet Service Providers (ISPs). BGP is also commonly used between ISPs and their large private clients to exchange routing information.

Network Access Layer

Address Resolution

- **ARP** - Address Resolution Protocol. Provides dynamic address mapping between an IPv4 address and a hardware address.

Note: You may see other documentation state that ARP operates at the Internet Layer (OSI Layer 3). However, in this course we state that ARP operates at the Network Access layer (OSI Layer 2) because its primary purpose is to discover the MAC address of the destination. A MAC address is a Layer 2 address.

Data Link Protocols

- **Ethernet** - Defines the rules for wiring and signaling standards of the network access layer.
- **WLAN** - Wireless Local Area Network. Defines the rules for wireless signaling across the 2.4 GHz and 5 GHz radio frequencies.

TCP/IP Protocol Suite in Three Real Scenarios

The TCP/IP suite is easiest to understand when you follow the traffic in the order it is actually used. In each scenario below, the outbound path shows how data is built and sent, and the inbound path shows how the response is received and interpreted.

Scenario 1: The First Connection - Bootstrap and Web

Outbound (Encapsulation)

1. **Ethernet** or **WLAN** provides the network access layer. It moves frames on a wired LAN or wireless LAN. **Ethernet** is chosen when the laptop is plugged in because it is stable and low latency. **WLAN** is chosen when mobility matters because it lets the laptop join the network without a cable.
2. **DHCPv4** or **DHCPv6** provides host configuration at the application layer. It gives the laptop an IP address, subnet information, gateway, and other settings. **DHCPv4** is chosen for IPv4 networks because it is the standard IPv4 bootstrap method. **DHCPv6** is chosen for

IPv6 networks when the organization wants managed IPv6 address assignment instead of self-configuration.

3. **SLAAC** provides IPv6 host configuration at the application layer. It lets the laptop form its own IPv6 address from router advertisements without waiting for a DHCPv6 server. **SLAAC** is chosen over **DHCPv6** when the network wants simple, low-management IPv6 addressing.
4. **ARP** operates at the network access layer. It resolves the next-hop IPv4 address into a MAC address so the laptop can deliver the first local frame. **ARP** is chosen because IPv4 needs a way to learn the Layer 2 destination before a packet can leave the LAN.
5. **DNS** operates at the application layer. It translates the website name into an IP address. **DNS** is chosen over hard-coding the address because users remember names, not numeric addresses, and services can move without changing the user workflow.
6. **HTTP** or **HTTPS** operates at the application layer. It requests the webpage from the server. **HTTPS** is chosen over **HTTP** when the page must be protected with encryption and server authentication. **HTTP** may still be used in lab or non-sensitive cases, but it does not protect data in transit.
7. **TCP** operates at the transport layer. It creates a reliable, ordered session for the browser request and the web response. **TCP** is chosen over **UDP** because web pages need confirmed delivery, retransmission, and in-order reassembly.
8. **IPv4** or **IPv6** operates at the internet layer. It addresses the packets end to end across routers. **IPv4** is chosen in older or mixed networks because it is still widely deployed. **IPv6** is chosen when the network wants a larger address space and cleaner modern addressing.

Inbound (De-encapsulation)

1. The web server returns data inside **TCP** segments, then **IPv4** or **IPv6** packets, then **Ethernet** or **WLAN** frames.
2. The laptop uses **TCP** to confirm, reorder, and reassemble the reply.
3. The laptop uses **DNS** results indirectly to match the response to the original site name.
4. If the laptop is on an IPv4 LAN, **ARP** may be used again to refresh or reuse the MAC address of the next hop.
5. The browser removes the application headers from **HTTP** or **HTTPS** and renders the page.

Why this chain works

The laptop cannot ask for a webpage until it has a usable address. That is why configuration comes first, name resolution comes second, and web delivery comes last.

Scenario 2: Corporate Communications - Messaging and Routing

Outbound (Encapsulation)

1. **SMTP** operates at the application layer. It sends the message from the mail client to the mail server, and later from one mail server to another. **SMTP** is chosen over **IMAP** and **POP3** for sending because those protocols retrieve mail, they do not deliver it.
2. **TCP** operates at the transport layer. It carries SMTP reliably so message text, attachments, and headers are not lost or reordered. **TCP** is chosen over **UDP** because email delivery must be dependable and acknowledged.
3. **IPv4** or **IPv6** operates at the internet layer. It carries the mail traffic across branch routers and provider networks. **IPv6** is chosen when the corporate backbone has modern addressing. **IPv4** is chosen when the branch, ISP, or legacy mail service still depends on it.
4. **OSPF** operates at the internet layer as an interior routing protocol. It moves routing information inside a corporate area or across branch sites. **OSPF** is chosen over **BGP** for internal routing because it is designed for the enterprise domain and area-based hierarchy.
5. **EIGRP** operates at the internet layer as an interior routing protocol. It can also move routes between branch sites inside a Cisco-focused enterprise. **EIGRP** is chosen over **OSPF** when the organization wants a Cisco-optimized design with composite metrics and it already standardizes on Cisco gear.
6. **BGP** operates at the internet layer as an exterior routing protocol. It exchanges routes between the enterprise edge and the ISP or between large autonomous systems. **BGP** is chosen over **OSPF** and **EIGRP** because inter-domain routing needs policy control, not just shortest-path convergence.
7. **REST APIs** operate at the application layer. They can be used by mail platforms, ticketing systems, or collaboration tools to create workflows such as routing alerts into mailboxes or pulling mailbox status into dashboards. **REST** is chosen over a direct mail protocol when the task is programmatic integration rather than email transport.

Inbound (De-encapsulation)

1. The destination mail server accepts the message, stores it, and may trigger downstream processing through **REST APIs**.
2. **BGP** or the internal routing protocol returns the route information needed for traffic to reach the remote office and to return the acknowledgment path.
3. **OSPF** or **EIGRP** keeps the corporate branches reachable by distributing the interior paths.
4. **TCP** confirms successful delivery of the SMTP session and ensures the mail was accepted.
5. The recipient later uses **IMAP** or **POP3** at the application layer to retrieve the message.

IMAP versus POP3

- **IMAP** is chosen when the user needs the same mailbox state on multiple devices because the message stays on the server.
- **POP3** is chosen when the user wants a simpler download model and local storage on one primary device.

Why this chain works

SMTP moves mail, IMAP or POP3 retrieves mail, and the routing protocols make sure the mail traffic reaches the right branch office and comes back. REST is adjacent support for automation, not the mail transport itself.

Scenario 3: Admin and Maintenance - Backend and Troubleshooting

Outbound (Encapsulation)

1. **SFTP** operates at the application layer. It securely uploads router images, configuration backups, or scripts. **SFTP** is chosen over **TFTP** when confidentiality, integrity, and authentication matter.
2. **TFTP** operates at the application layer. It can transfer firmware or recovery files with very little overhead. **TFTP** is chosen over **SFTP** when the technician needs simplicity, small footprint, or a recovery path on a constrained device, even though it is not secure.
3. **UDP** operates at the transport layer for **TFTP**. It carries the file blocks without connection setup or acknowledgments. **UDP** is chosen over **TCP** because TFTP values low overhead and simple request-response behavior more than reliability features.
4. **NAT** operates at the internet layer for IPv4 deployments. It translates private IPv4 addresses from the management network into a public or shared address when traffic must cross an edge boundary. **NAT** is chosen because many private management networks cannot be routed directly on the public internet.
5. **ICMPv4** or **ICMPv6** operates at the internet layer. It checks reachability with echo request and echo reply messages and reports delivery problems. **ICMPv4** is chosen for IPv4 troubleshooting, while **ICMPv6** is chosen for IPv6 troubleshooting because each one matches its address family.
6. **ICMPv6 ND** operates at the internet layer. It performs IPv6 neighbor discovery and duplicate address detection. **ICMPv6 ND** is chosen over **ARP** in IPv6 networks because IPv6 uses Neighbor Discovery instead of ARP for local address resolution.

Inbound (De-encapsulation)

1. The router receives the file blocks from **SFTP** or **TFTP** and reconstructs the image or configuration.
2. The router or target host returns **ICMPv4** or **ICMPv6** replies to confirm reachability.
3. **ICMPv6 ND** confirms the local IPv6 neighbor and helps the technician verify that the device can learn the next-hop link-layer information.
4. **NAT** reverses translation on return traffic so the technician sees replies mapped back to the original private source.
5. The technician checks whether the file transfer and connectivity test completed successfully.

Why this chain works

Maintenance traffic often needs either secure transfer or low-overhead recovery transfer. Connectivity checks use control-plane messaging, not application data, because the technician is validating reachability before trusting the device.

Protocol Interaction Map

- **Ethernet/WLAN** carries the local frame that starts every scenario.
- **DHCPv4/DHCPv6/SLAAC** gives the host an address before anything else can work.
- **ARP** or **ICMPv6 ND** maps the next hop at Layer 2 so the local network can forward traffic.
- **IPv4/IPv6** moves the packet between networks and routers.
- **TCP** gives reliable delivery for web, email submission, and secure file transfer.
- **UDP** gives lightweight delivery for **TFTP** and other low-overhead exchanges.
- **DNS** turns names into addresses so applications know where to send traffic.
- **HTTP/HTTPS** delivers web content after addressing and transport are in place.
- **SMTP** sends mail, while **IMAP/POP3** retrieve it later.
- **OSPF** and **EIGRP** move routes inside the enterprise, while **BGP** moves routes between enterprises and ISPs.
- **REST APIs** automate systems around the communication stack, but they still depend on **HTTP/HTTPS**, **TCP**, and **IP** underneath.
- **SFTP** and **TFTP** move maintenance files, with **SFTP** preferred for security and **TFTP** preferred for minimal overhead.
- **ICMPv4/ICMPv6** verifies reachability and reports errors, which helps confirm that the rest of the stack is healthy.

The practical dependency chain is simple: addressing enables routing, routing enables transport, transport enables applications, and control protocols verify that the path still works.

5.2.4 Message Formatting and Encapsulation

When a message is sent from source to destination, it must use a specific format or structure. Message formats depend on the type of message and the channel that is used to deliver the message.

Analogy: Letter Formatting

A common example of requiring the correct format in human communications is when sending a letter. An envelope has the address of the sender and receiver, each located at the proper place on the envelope. If the destination address and formatting are not correct, the letter is not delivered.

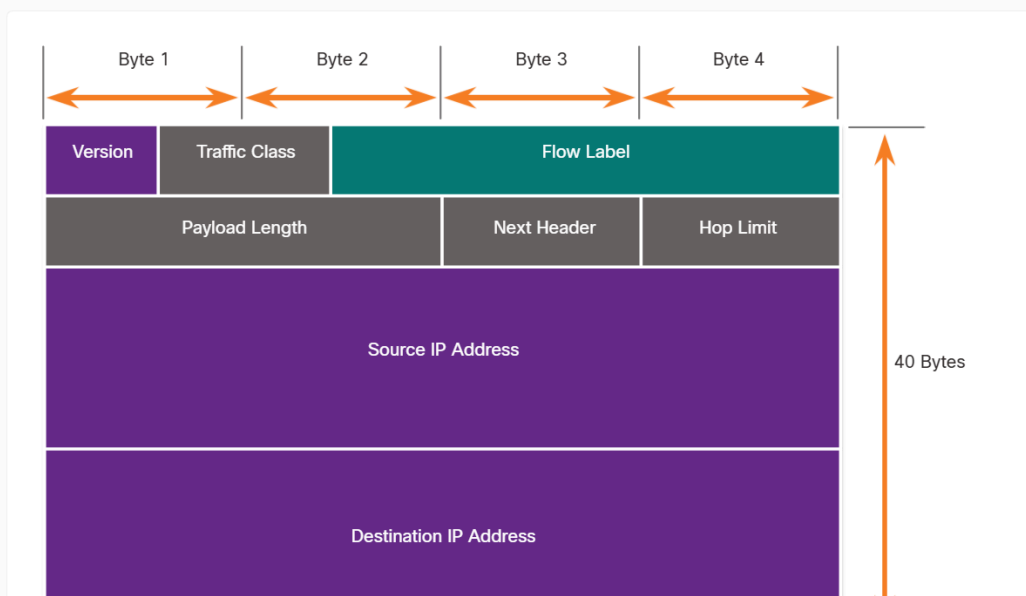
The process of placing one message format (the letter) inside another message format (the envelope) is called **encapsulation**. **De-encapsulation** occurs when the process is reversed by the recipient and the letter is removed from the envelope.

Network

Similar to sending a letter, a message that is sent over a computer network follows specific format rules for it to be delivered and processed.

Internet Protocol (IP) is a protocol with a similar function to the envelope example. In the figure, the fields of the Internet Protocol version 6 (IPv6) packet identify the source of the packet and its destination. IP is responsible for sending a message from the message source to destination over one or more networks.

Note: The fields of the IPv6 packet are discussed in detail in another module.



5.2.5 Message Size

Another rule of communication is message size.

Analogy: Face-to-Face Communication

When people communicate with each other, the messages that they send are usually broken into smaller parts or sentences. These sentences are limited in size to what the receiving person can process at one time. It also makes it easier for the receiver to read and comprehend.

Network Message Size

Likewise, when a long message is sent from one host to another over a network, it is necessary to break the message into smaller pieces. The rules that govern the size of the pieces, or frames, communicated across the network are very strict. They can also be different, depending on the channel used. Frames that are too long or too short are not delivered.

The size restrictions of frames require the source host to break a long message into individual pieces that meet both the minimum and maximum size requirements. The long message will be sent in separate frames, with each frame containing a piece of the original message. Each frame will also have its own addressing information. At the receiving host, the individual pieces of the message are reconstructed into the original message.

5.2.6 Message Timing

Message timing is also very important in network communications. Message timing includes the following:

Flow Control - This is the process of managing the rate of data transmission. Flow control defines how much information can be sent and the speed at which it can be delivered. For example, if one person speaks too quickly, it may be difficult for the receiver to hear and understand the message. In network communication, there are network protocols used by the source and destination devices to negotiate and manage the flow of information.

Response Timeout - If a person asks a question and does not hear a response within an acceptable amount of time, the person assumes that no answer is coming and reacts accordingly. The person may repeat the question or instead, may go on with the conversation. Hosts on the network use network protocols that specify how long to wait for responses and what action to take if a response timeout occurs.

Access Method - This determines when someone can send a message. When a device wants to transmit on a wireless LAN, it is necessary for the WLAN network interface card (NIC) to determine whether the wireless medium is available. A collision of information occurs when two devices try to send at the same time, and it is necessary for the two to back off and start again.

5.2.7 Unicast, Multicast, and Broadcast

A message can be delivered in different ways. Sometimes, a person wants to communicate information to a single individual. At other times, the person may need to send information to a group of people at the same time, or even to all people in the same area.

Hosts on a network use similar delivery options to communicate. These methods of communication are called unicast, multicast, and broadcast.

Unicast

- A one-to-one delivery option referred to as unicast, meaning there is only a single destination for the message.

Multicast

- When a host needs to send messages using a one-to-many delivery option, it is referred to as multicast.

Broadcast

- If all hosts on the network need to receive the message at the same time, a broadcast may be used. Broadcasting represents a one-to-all message delivery option.
-

5.2.8 The Benefits of Using a Layered Model

You cannot actually watch real packets travel across a real network the way you can watch the components of a car being put together on an assembly line. So, it helps to have a way of thinking about a network so that you can imagine what is happening. A model is useful in these situations.

Complex concepts such as how a network operates can be difficult to explain and understand. For this reason, a layered model is used to modularize the operations of a network into manageable layers.

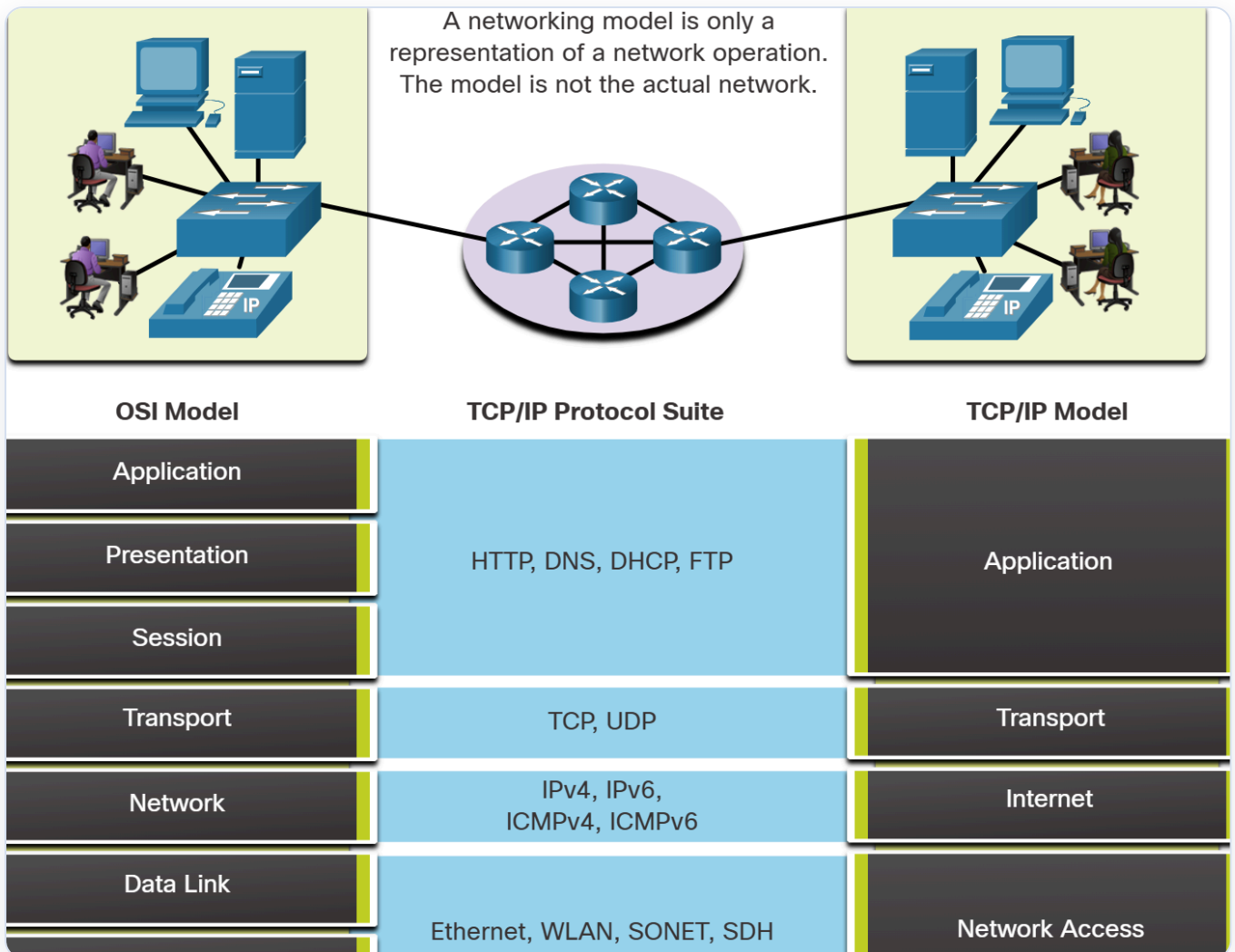
Benefits of Using a Layered Model:

- Assisting in protocol design because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below
- Fostering competition because products from different vendors can work together
- Preventing technology or capability changes in one layer from affecting other layers above and below
- Providing a common language to describe networking functions and capabilities

Two Layered Models:

1. Open System Interconnection (OSI) Reference Model

2. TCP/IP Reference Model



5.2.9 The OSI Reference Model

The OSI reference model provides an extensive list of functions and services that can occur at each layer. This type of model provides consistency within all types of network protocols and services by describing what must be done at a particular layer, but not prescribing how it should be accomplished.

It also describes the interaction of each layer with the layers directly above and below. The TCP/IP protocols discussed in this course are structured around both the OSI and TCP/IP models.

OSI Model Layer	Description
7 - Application	The application layer contains protocols used for process-to-process communications.
6 - Presentation	The presentation layer provides for common representation of the data transferred between application layer services.
5 - Session	The session layer provides services to the presentation layer to organize its dialogue and to manage data exchange.
4 - Transport	The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
3 - Network	The network layer provides services to exchange the individual pieces of data over the network between identified end devices.
2 - Data Link	The data link layer protocols describe methods for exchanging data frames between devices over a common media.
1 - Physical	The physical layer protocols describe the mechanical, electrical, functional, and procedural means to activate, maintain, and de-activate physical connections for a bit transmission to and from a network device.

Note: Whereas the TCP/IP model layers are referred to only by name, the seven OSI model layers are more often referred to by number rather than by name. For instance, the physical layer is referred to as Layer 1 of the OSI model, data link layer is Layer 2, and so on.

5.2.10 The TCP/IP Protocol Model

The TCP/IP protocol model for internetwork communications was created in the early 1970s and is sometimes referred to as the internet model. This type of model closely matches the structure of a particular protocol suite. The TCP/IP model is a protocol model because it describes the functions that occur at each layer of protocols within the TCP/IP suite. TCP/IP is also used as a reference model. The table shows details about each layer of the OSI model.

TCP/IP Model Layer	Description
4 - Application	Represents data to the user, plus encoding and dialog control.
3 - Transport	Supports communication between various devices across diverse networks.
2 - Internet	Determines the best path through the network.
1 - Network Access	Controls the hardware devices and media that make up the network.

The definitions of the standard and the TCP/IP protocols are discussed in a public forum and defined in a publicly available set of IETF request for comment (RFC) documents. An RFC is authored by networking engineers and sent to other IETF members for comments.

The functionality of each layer and the relationship between layers will become more evident throughout this course as the protocols are discussed in more detail.

5.3 Data Encapsulation

5.3.1 Segmenting Messages

Knowing the OSI reference model and the TCP/IP protocol model will come in handy when you learn about how data is encapsulated as it moves across a network. It is not as simple as a physical letter being sent through the mail system.

In theory, a single communication, such as a video or an email message with many large attachments, could be sent across a network from a source to a destination as one massive, uninterrupted stream of bits. However, this would create problems for other devices needing to use the same communication channels or links. These large streams of data would result in significant delays. Further, if any link in the interconnected network infrastructure failed during the transmission, the complete message would be lost and would have to be retransmitted in full.

A better approach is to divide the data into smaller, more manageable pieces to send over the network. Segmentation is the process of dividing a stream of data into smaller units for transmissions over the network. Segmentation is necessary because data networks use the TCP/IP protocol suite send data in individual IP packets. Each packet is sent separately, similar to sending a long letter as a series of individual postcards. Packets containing segments for the same destination can be sent over different paths.

This leads to segmenting messages having two primary benefits:

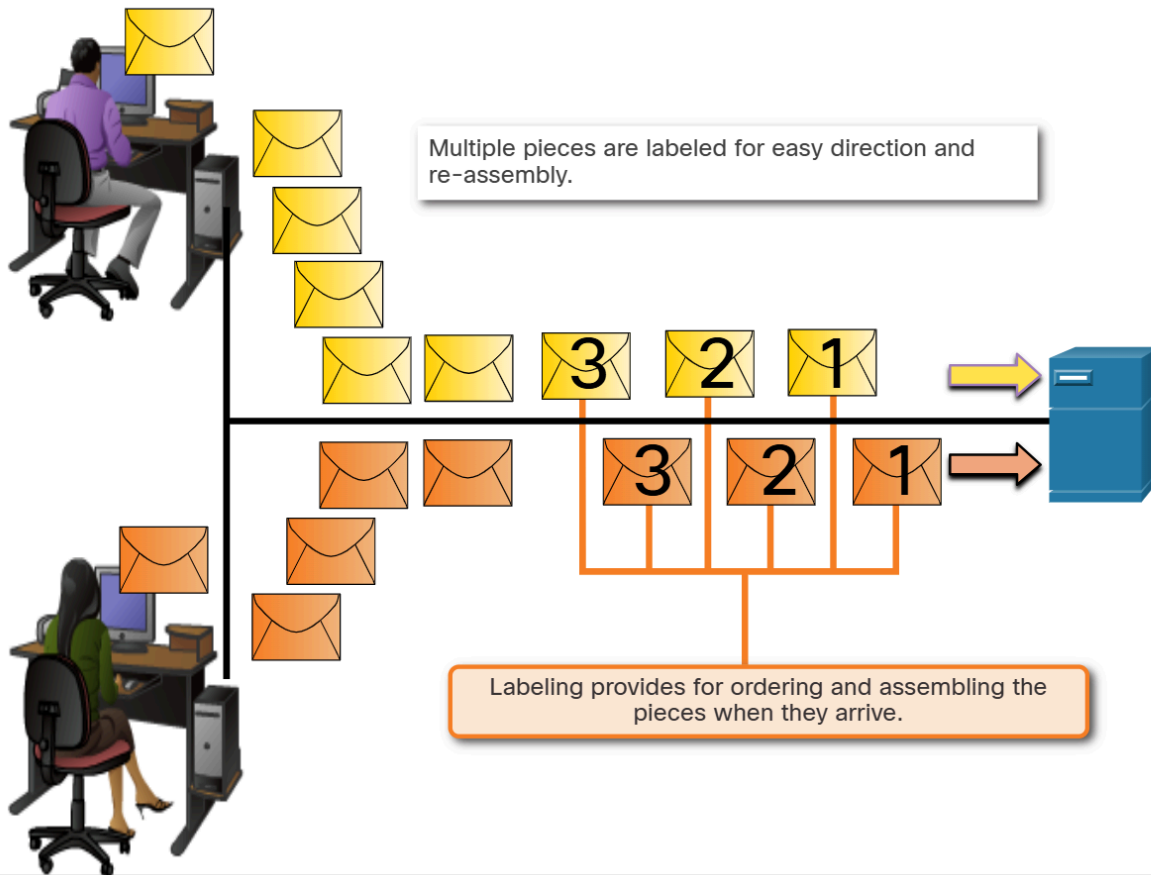
- **Increases speed** - Because a large data stream is segmented into packets, large amounts of data can be sent over the network without tying up a communications link. This allows many different conversations to be interleaved on the network called multiplexing.
- **Increases efficiency** - If a single segment fails to reach its destination due to a failure in the network or network congestion, only that segment needs to be retransmitted instead of resending the entire data stream.

Click each button in the figure to view the animations of segmentation and multiplexing.

5.3.2 Sequencing

The challenge to using segmentation and multiplexing to transmit messages across a network is the level of complexity that is added to the process. Imagine if you had to send a 100-page letter, but each envelope could only hold one page. Therefore, 100 envelopes would be required, and each envelope would need to be addressed individually. It is possible that the 100-page letter in 100 different envelopes arrives out-of-order. Consequently, the information in the envelope would need to include a sequence number to ensure that the receiver could reassemble the pages in the proper order.

In network communications, each segment of the message must go through a similar process to ensure that it gets to the correct destination and can be reassembled into the content of the original message, as shown in the figure. TCP is responsible for sequencing the individual segments.

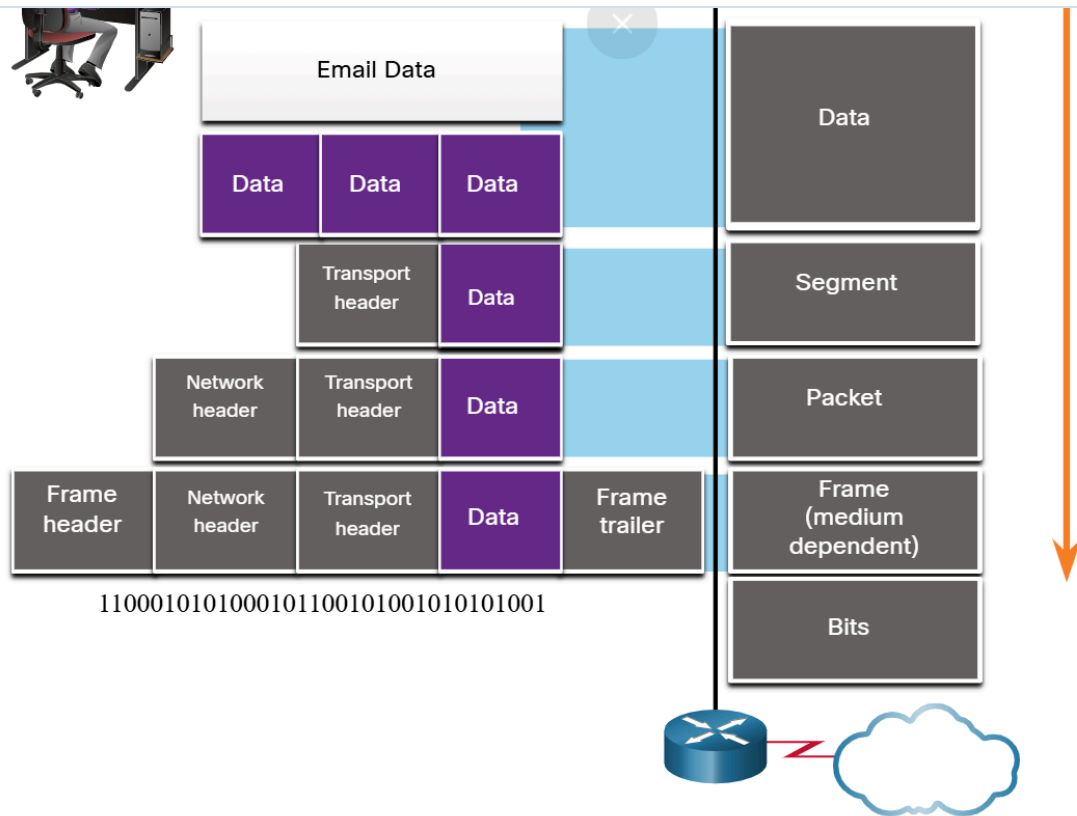


5.3.3 Protocol Data Units

As application data is passed down the protocol stack on its way to be transmitted across the network media, various protocol information is added at each level. This is known as the encapsulation process.

Note: Although the UDP PDU is called datagram, IP packets are sometimes also referred to as IP datagrams.

The form that a piece of data takes at any layer is called a protocol data unit (PDU). During encapsulation, each succeeding layer encapsulates the PDU that it receives from the layer above in accordance with the protocol being used. At each stage of the process, a PDU has a different name to reflect its new functions. Although there is no universal naming convention for PDUs, in this course, the PDUs are named according to the protocols of the TCP/IP suite. The PDUs for each form of data are shown in the figure.



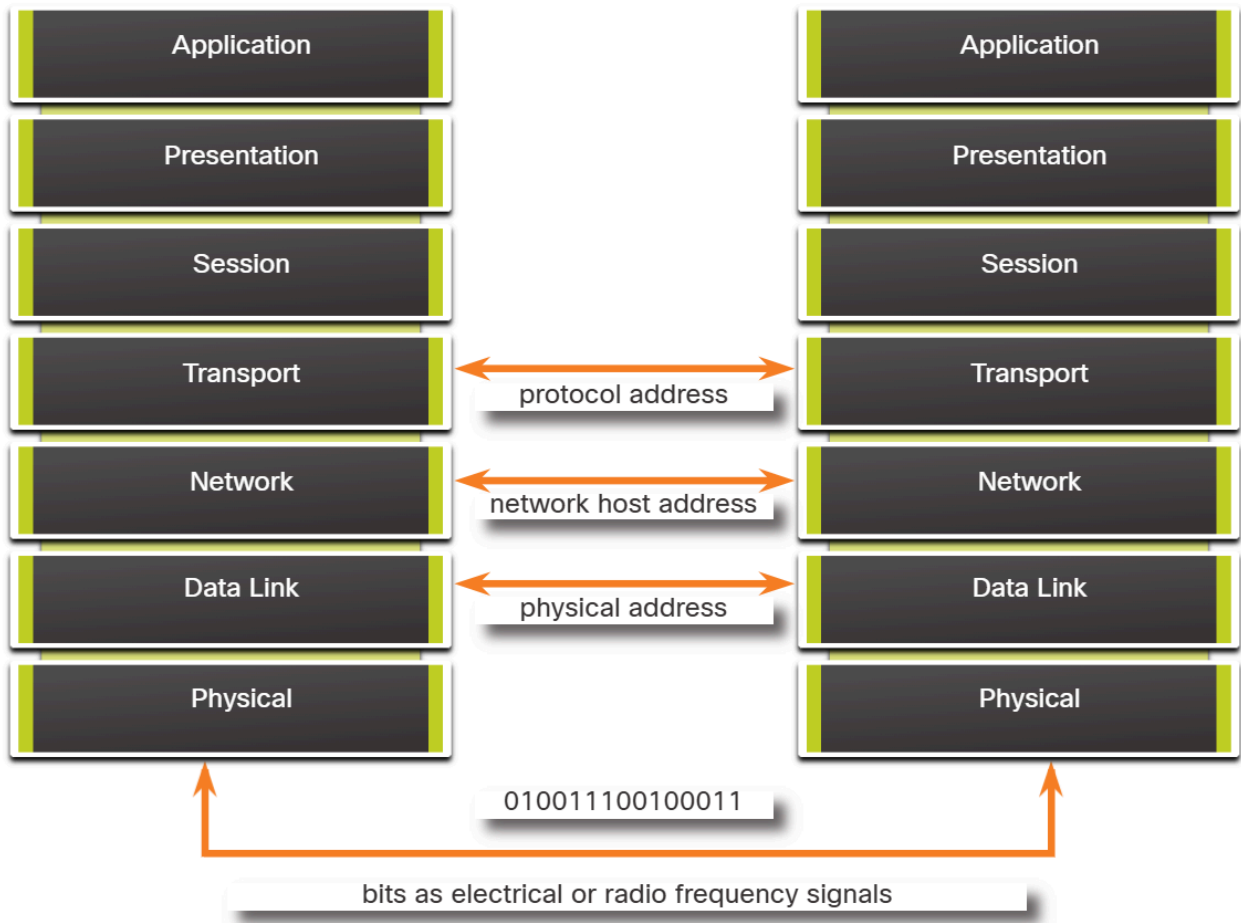
- *Data* - The general term for the PDU used at the application layer
- *Segment* - Transport layer PDU
- *Packet* - Network layer PDU
- *Frame* - Data Link layer PDU
- *Bits* - Physical layer PDU used when physically transmitting data over the medium

Note: If the Transport header is TCP, then it is a segment. If the Transport header is UDP then it is a datagram.

5.3.4 Three Addresses

Network protocols require that addresses be used for network communication. Addressing is used by the client to send requests and other data to a server. The server uses the client's address to return the requested data to the client that requested it.

The OSI transport, network, and data link layers all use addressing in some form. The transport layer uses protocol addresses in the form of port numbers to identify network applications that should handle client and server data. The network layer specifies addresses that identify the networks that clients and servers are attached to and the clients and servers themselves. Finally, the data link layer specifies the devices on the local LAN that should handle data frames. All three addresses are required for client-server communication, as shown in the figure.



5.3.5 Encapsulation Example

When messages are being sent on a network, the encapsulation process works from top to bottom. At each layer, the upper layer information is considered data within the encapsulated protocol. For example, the TCP segment is considered data within the IP packet.

You saw this animation previously in this module. This time, click Play and focus on the encapsulation process as a web server sends a web page to a web client.

5.3.6 De-encapsulation Example

This process is reversed at the receiving host and is known as de-encapsulation. De-encapsulation is the process used by a receiving device to remove one or more of the protocol headers. The data is de-encapsulated as it moves up the stack toward the end-user application.

You saw this animation previously in this module. This time, click Play and focus on the de-encapsulation process.

5.3.7 Lab - Introduction to Wireshark

Wireshark is a software protocol analyzer, or "packet sniffer" application, used for network troubleshooting, analysis, software and protocol development, and education. Wireshark is used throughout the course to demonstrate network concepts. In this lab, you will use Wireshark to capture and analyze network traffic.

 Introduction to Wireshark

Practical Lab Addendum: Packet Tracer P2P Encapsulation

The Packet Tracer lab below turns the theory above into a direct observation exercise. A PC at 192.168.1.1 was connected to a Server at 192.168.1.2 so the OSI stack could be watched from link setup through the final ICMP reply.

Theoretical Concept vs. Practical Evidence

OSI Layer	Theory	Practical evidence from the lab	Exam cue
Layer 1 - Physical	Signals must travel over the medium before any data can move	Red triangles appeared when a Straight-Through cable was used between like devices; swapping to a Copper Cross-Over cable restored green link indicators	Fix Layer 1 first when the link will not come up
Layer 2 - Data Link	Frames use MAC addresses for local delivery	A Gratuitous ARP Request was seen as a broadcast to FFFF.FFFF.FFFF, and <code>arp -a</code> confirmed the MAC-to-IP mapping	MAC addresses solve local delivery on the LAN
Layer 3 - Network	IP handles logical source and destination addressing	In the ICMP Echo Reply PDU details, the source and destination IP addresses swapped as the reply returned from 192.168.1.2 to 192.168.1.1	IP addresses identify the endpoints end to end
Layer 4 - Transport	Transport controls end-to-end conversation behavior	Ping did not use TCP; the successful test showed that reachability was proven with ICMP rather than a transport session	Do not confuse ping with TCP-based application traffic
Layer 5 - Session	Session logic keeps a conversation organized	Simulation Mode showed the request and reply as separate stages of one conversation rather than one continuous blob of data	Think of this as the conversation state around the exchange
Layer 6 - Presentation	Data is formatted for interpretation	The packet became easier to read as headers were added and stripped during Simulation Mode	Presentation is where data becomes usable to the next layer
Layer 7 - Application	User-facing data begins and ends the communication chain	The ping command initiated the request, and the final reply confirmed the host could be reached	Start at the application intent, then trace downward

Lab Summary

Lab Report: P2P Encapsulation & Connectivity

Objective: To visualize the 7-layer OSI encapsulation process using a direct connection.

Hardware Setup: 1x PC, 1x Server, 1x Copper Cross-Over Cable (FastEthernet0 interfaces).

Logic Applied:

- Layer 1: Verified physical link lights. Learned that like devices require crossed TX/RX pairs.
- Layer 2: Observed PDU formation. The NIC added the frame header containing the physical MAC address. Verified local mapping via the `arp -a` command.
- Layer 3: Manually assigned IPv4 addresses. Verified end-to-end connectivity using the `ping` command.

Results: Successfully captured a 4-packet ICMP exchange.

Sent: 4, Received: 4, Loss: 0%.

Observation: In Simulation Mode, the PDU headers were stripped during de-encapsulation as the packet moved up the stack at the receiver and re-added during encapsulation for the reply.

Key Takeaways for Exam Prep

- Layer 2 is physical addressing: it uses MAC addresses for local delivery on the LAN.
- Layer 3 is logical addressing: it uses IP addresses for end-to-end source and destination identification.
- ARP bridges the two layers by mapping an IPv4 address to a MAC address.
- A broadcast ARP frame uses FFFF.FFFF.FFFF, while the IP addresses still stay focused on the endpoints.
- If the link shows red triangles, troubleshoot the cable and Layer 1 before checking IP settings.
- Ping success proves Layer 3 reachability, not application reliability or TCP session behavior.

Lab 02: Layer 2 Intelligence & Switch Logic

Methodology

The lab moved from a peer-to-peer cross-over connection to a star topology using a Cisco 2960 switch. The cross-over cable was appropriate for the original host-to-host test because like interfaces require crossed transmit and receive pairs.

When the star topology was introduced, straight-through cables were used from each host to the switch. That choice is correct for host-to-node connections because the switch is the termination point for the access link and the cable does not need to cross the pairs manually.

The switch became the central Layer 2 learning device and the forwarding decision point for the new topology.

PDU Analysis Table

Layer	Data Type	Outbound / Inbound Observation	Source / Destination Address Swap
Layer 2 - Data Link	Ethernet II	The frame carried local MAC addressing between the PC and the switch	The return frame used the host MAC as the local destination for delivery back to the sender
Layer 3 - Network	IP	The packet preserved end-to-end logical addressing for the ICMP exchange	The source and destination IP addresses reversed in the Echo Reply
Layer 3 - Network	ICMP	The Echo Request triggered reachability testing and the Echo Reply confirmed connectivity	The ICMP control message returned to the original host along the reverse path

Dynamic Learning Observation

The `show mac address-table` command showed the switch beginning with an empty learning state and then populating its table after the first ICMP Echo Request. This demonstrated how the switch builds its Layer 2 "brain" from observed source MAC addresses on active ports.

Before the first successful exchange, the MAC table contained no meaningful learned entries for the test hosts. After the first frame crossed the switch, the source MAC was associated with the ingress port and the switch could forward later frames with much higher confidence.

Troubleshooting Section

The initial symptom was `Request Timed Out`. The issue was tied to the ARP resolution phase and the timing of Packet Tracer Simulation Mode, not to a broken IP configuration. The packet flow had not yet completed the local address resolution needed for the ICMP exchange.

The resolution was to switch from Simulation Mode to Realtime Mode. That allowed ARP resolution and switch learning to complete normally, after which the ping succeeded without further intervention.

Research Application

Layer 2 MAC tables are the precursor to SDN Flow Tables because both capture learned forwarding intelligence that can be observed and managed. A traditional switch uses local MAC learning to forward frames in hardware, while an SDN controller can build higher-level flow policies from that same kind of forwarding behavior.

This is especially relevant to Susan-Sense, where rural hazard detection depends on low-latency switching at the edge. Camera, sensor, and alert traffic must move quickly and predictably so events can be detected and reported without delay. In that context, Layer 2 transparency is a prerequisite for timely situational awareness, controller visibility, and reliable automation.

Web Request Lifecycle Through the OSI Model

The scenario is simple: a user types `https://www.google.com` into a browser and presses Enter. The browser must first turn the name into an address, then build a secure web session, then send the request across the network, and finally receive and render the response.

Phase 1: The Request - Client Outbound

Layer 7 - Application

- **PDU:** Data
- **Action:** The browser creates the web request and asks for the site by name.
- **Protocols:** **DHCP**, **DNS**, and **HTTPS**.
- **Protocol logic:** **DHCP** is used first during initial setup so the client can get an IP address, gateway, and DNS server. **DNS** is used because the browser knows a name, not the server's numeric address. **HTTPS** is used instead of plain **HTTP** because the request and response must be protected with encryption and server authentication.

Layer 6 - Presentation

- **PDU:** Data
- **Action:** The client prepares the payload format and encrypts application data.
- **Protocols:** **TLS/SSL**, **ASCII**, and **UTF-8**.
- **Protocol logic:** **TLS/SSL** is used because web traffic must be encrypted before it leaves the client. **ASCII** and **UTF-8** define how text characters are encoded into bytes so both sides interpret headers and content consistently. **UTF-8** is preferred over plain **ASCII** for modern web content because it supports a much larger character set.

Layer 5 - Session

- **PDU:** Data
- **Action:** The client establishes and maintains the logical session for the conversation.
- **Protocols:** **NetBIOS** and session control logic.
- **Protocol logic:** **NetBIOS** is a classic session-layer example used to manage naming and session coordination in legacy environments. It is not the primary protocol for modern web browsing, but it represents the session-layer idea of opening, maintaining, and closing a conversation. For a browser session, the session role is usually handled by the application and TLS stack rather than by a separate visible session protocol.

Layer 4 - Transport

- **PDU:** Segment
- **Action:** The request is placed into a **TCP** segment and prepared for reliable delivery.
- **Protocols:** **TCP**.
- **Protocol logic:** **TCP** is chosen instead of **UDP** because a web request needs reliability, ordering, and retransmission. The browser depends on a confirmed connection before it can trust the page data. The **3-way handshake** creates the session: SYN, SYN-ACK, and ACK.

Layer 3 - Network

- **PDU:** Packet
- **Action:** The segment is wrapped in an **IPv4** or **IPv6** packet so it can be routed across networks.
- **Protocols:** **IPv4**, **IPv6**, and **ICMP** for connectivity checks.
- **Protocol logic:** **IPv4** is used in many existing networks, while **IPv6** is used where the organization wants a larger address space and modern addressing. **ICMP** is not used for the web data path itself, but it is used to verify connectivity and report errors if the path fails.

Before the packet leaves the client LAN, **ARP** is used when the client is on IPv4 to find the MAC address of the default gateway. **ARP** is chosen because the packet cannot be placed on the local wire without a Layer 2 destination.

Layer 2 - Data Link

- **PDU:** Frame
- **Action:** The packet is wrapped in an **Ethernet** or **WLAN** frame.
- **Protocols:** **Ethernet** and **WLAN**.

- **Protocol logic: Ethernet** is used on wired links because it is efficient and deterministic on copper or fiber. **WLAN** is used on wireless links because the client is moving through radio space instead of a cable. The frame provides local delivery to the next hop, not end-to-end delivery.

Layer 1 - Physical

- **PDU:** Bits
- **Action:** The frame is converted into electrical signals, light pulses, or radio waves.
- **Protocols:** Physical signaling rules for copper, fiber, or radio.
- **Protocol logic:** This layer does not decide what the message means. It only carries bits across the medium as voltage changes, light changes, or wireless energy patterns.

Phase 2: The Journey - Router or Middlebox

The router only processes up to Layer 3. It removes the Layer 2 frame, examines the Layer 3 packet, and decides where to send it next. It does not inspect the web page content at Layer 7 for basic forwarding. If the device is acting as a firewall, proxy, or SSL inspection middlebox, it may look higher, but a normal router forwards based on the network layer.

- It reads the **IP** header to determine the next hop.
- It may decrement the hop count or TTL so packets cannot loop forever.
- It chooses a new outgoing **Ethernet/WLAN** frame for the next link.
- It uses **ARP** again on IPv4 networks if it needs the next-hop MAC address.
- It may let **ICMP** errors travel back if the route fails or a network is unreachable.

Phase 3: The Arrival - Server Inbound

The server receives the traffic in reverse order, from Layer 1 up to Layer 7.

Layer 1 - Physical

- **PDU:** Bits
- **Action:** The server receives light, electrical, or radio signals and converts them back into a frame.

Layer 2 - Data Link

- **PDU:** Frame
- **Action:** The server checks the frame, verifies the destination MAC address, and removes the Layer 2 header and trailer.

Layer 3 - Network

- **PDU:** Packet
- **Action:** The server verifies the **IPv4** or **IPv6** header and confirms the packet is addressed correctly.
- **Protocol logic:** **IP** is used because the server must know the end-to-end source and destination before it can continue upward.

Layer 4 - Transport

- **PDU:** Segment
- **Action:** The server processes the **TCP** segment, confirms sequence and acknowledgment state, and hands the data to the correct process.
- **Protocol logic:** **TCP** is used because it guarantees the server receives the full request in order. **UDP** would not provide that reliability for the web transaction.

Layer 5 - Session

- **PDU:** Data
- **Action:** The server maintains the logical session and associates the request with the active conversation.
- **Protocol logic:** Session handling keeps the user interaction coherent across multiple exchanges, even though modern web stacks often blend this function into higher-layer logic.

Layer 6 - Presentation

- **PDU:** Data
- **Action:** The server decrypts the payload and interprets the character encoding.
- **Protocol logic:** **TLS/SSL** is removed so the server can read the request securely. **UTF-8** or **ASCII** is decoded so the server can interpret the text correctly.

Layer 7 - Application

- **PDU:** Data
- **Action:** The web server understands the request, locates the content, and generates the response.
- **Protocol logic:** **HTTPS** is the application-facing service that returns the page content securely. The server may also use **DNS** information internally to support name-based services, but the main application behavior here is responding to the browser request.

Phase 4: The Response - Server to Client

The response uses the same stack in reverse.

- The server creates application data at Layer 7, then formats and encrypts it at Layer 6 with **TLS/SSL**.
- The logical session is maintained at Layer 5.
- The response is carried by **TCP** at Layer 4 because the browser still needs reliable delivery.
- The data is packaged into **IPv4** or **IPv6** packets at Layer 3.
- The packets are delivered inside **Ethernet** or **WLAN** frames at Layer 2.
- The physical medium carries the bits at Layer 1.
- The client then de-encapsulates upward, verifies the **TCP** stream, decrypts the **TLS/SSL** payload, and renders the page.

Header Summary Table

OSI Layer	PDU Name	Main Header Added	What It Carries
Layer 7 - Application	Data	HTTP or HTTPS request data	The browser's request for the webpage
Layer 6 - Presentation	Data	TLS/SSL and encoding information	Encryption and character formatting
Layer 5 - Session	Data	Session control information	Conversation state and dialog control
Layer 4 - Transport	Segment	TCP header	Ports, sequencing, acknowledgment, and reliability
Layer 3 - Network	Packet	IPv4 or IPv6 header	Source and destination IP addressing and routing
Layer 2 - Data Link	Frame	Ethernet or WLAN header and trailer	MAC addresses and local-link delivery
Layer 1 - Physical	Bits	No header, only signaling	Electrical, light, or radio bit transmission

Quick Takeaway

The browser does not send a webpage directly. It sends data that gets wrapped by each lower layer until the network can move it reliably across the path. On the return path, the server and

the client remove those layers in reverse order until the browser can display the page.